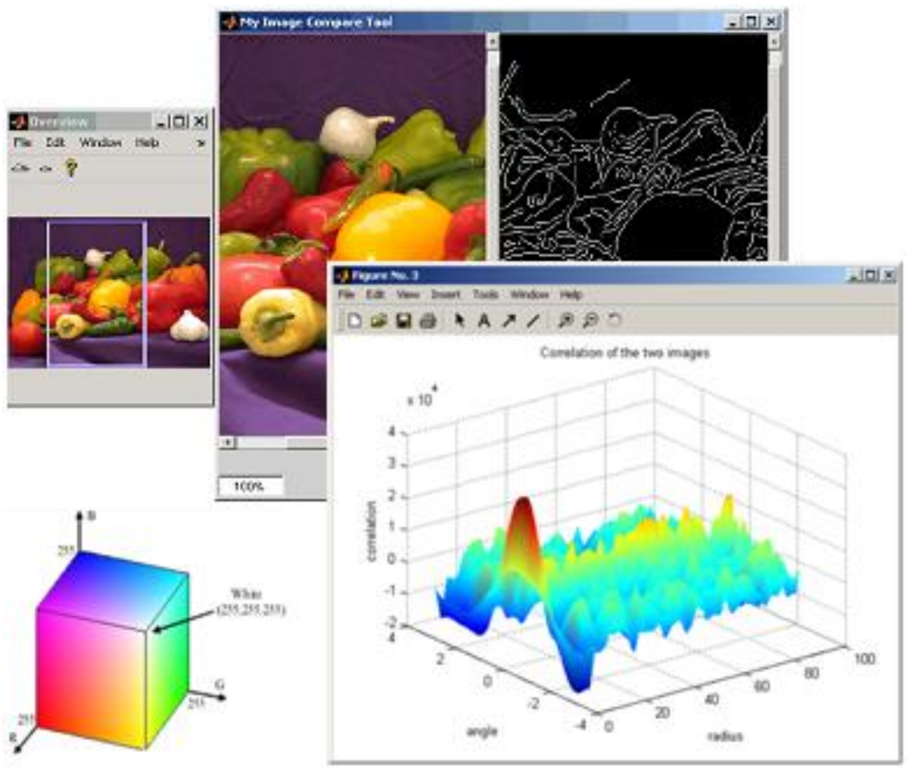
Digital Image Processing

HISTOGRAM PROCESSING
(UNIVERSITY QUESTIONS)



Dr. Ajay Kumar Mahato
Assistant Professor
ECE Department
GLA University Mathura

Histogram Plotting

Numerical Examples

Ques 2: Construct the histogram for the image shown below:

GLA University, Carry Exam, 2023-24

0	0	0	4	6
3	3	3	2	1
5	5	3	2	2
2	2	0	1	1
4	4	3	1	3

Histogram Equalization

Numerical Examples

TYPE 1

Ques 1: Suppose that a 3-bit image (L = 8) of size 64 × 64 pixels (MN = 4096) has the intensity distribution shown in following table. Get the histogram equalization transformation function and give the $p(s_k)$ for each s_k

r_k	0	1	2	3	4	5	6	7
n_k	790	1023	850	656	329	245	122	81

Numerical Examples

TYPE 1

Ques: Equalize the following histogram of a 8×8 image

Gray Levels	0	1	2	3	4	5	6	7
No of Pixels	8	10	10	2	12	16	4	2

GLA University, Mid term Examination, Odd Semester 2022-23

Numerical Example

TYPE 2

Ques2: Apply the histogram Equalization technique to the following image

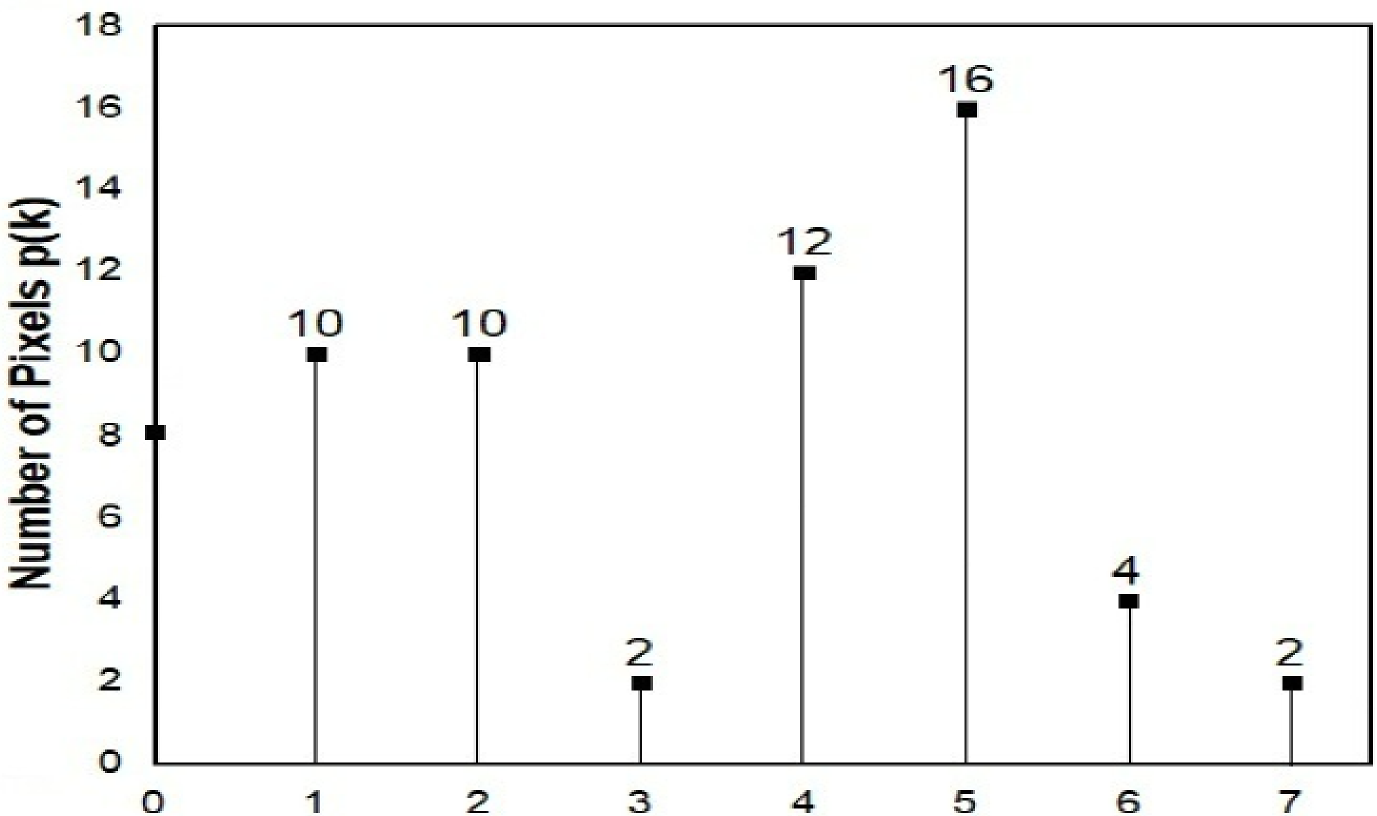
$$f(x, y) = \begin{array}{|c|c|c|c|} \hline 1 & 2 & 3 & 4 \\ \hline 5 & 5 & 6 & 6 \\ \hline 6 & 7 & 6 & 6 \\ \hline 6 & 7 & 2 & 3 \\ \hline \end{array}$$

Numerical Examples

Ques 8: Perform Histogram Equalisation for 8 × 8 eight level image described in Figure. Also, construct the histogram of the equalized image.

GLA University, Mid Term Examination, Odd Semester 2024-25

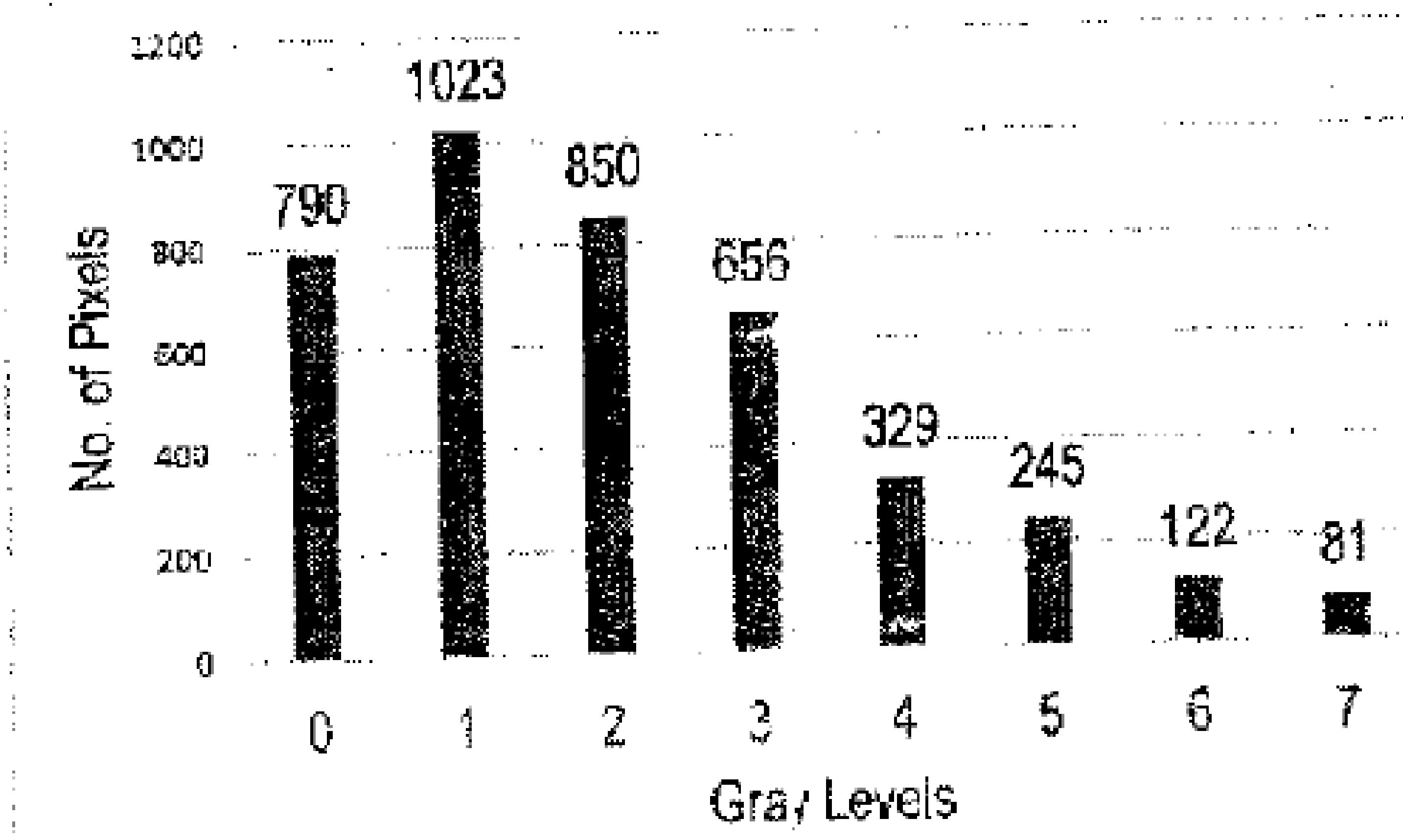
TYPE 3



Numerical Examples

Ques: Following is the histogram of a 3-bit image. Plot the histogram of the Equalised image. Show all the calculation in the tabular format.

GLA University, Mid Term Examination, Odd Semester 2023-24



Numerical Examples

Ques: Histogram of a 3bit/pixel square image is given in figure-1

GLA University, Mid Term Examination, Odd Semester 2024-25

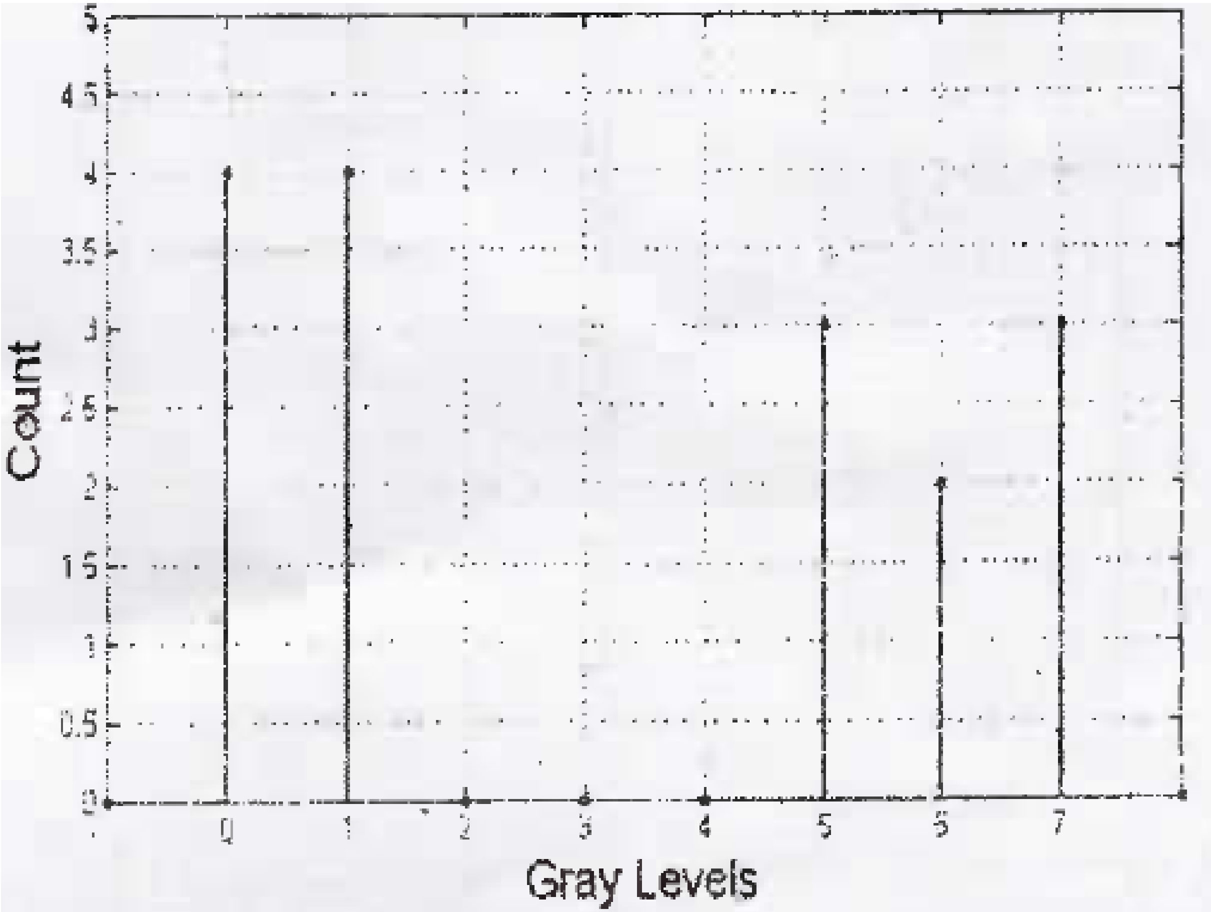


Figure-1 Histogram

Numerical Examples

a. Find out the size of the image and total number of pixels

b. What would be the new histogram if we make MSB (Most Significant bit) equal to 1.

c. Compute threshold (T) such that T=m, where m is the mean gray level of original image. Apply this threshold and draw new histogram for output binary image.

Histogram Matching (Specification)

Numerical Examples

TYPE 1

Ques1: Perform histogram specification on the 8 × 8, eight level image described in Table below

Table I: Pixel distribution of the image

i/p gray level (r_k)	0	1	2	3	4	5	6	7
No. of pixel (n_k)	8	10	10	2	12	16	4	2

Table II: Target Histogram

i/p gray level (r_k)	0	1	2	3	4	5	6	7
No. of pixel (n_k)	0	0	0	0	20	20	16	8

Numerical Examples

Ques: Perform the histogram specification on 8×8, eight gray level image histogram described below:

GLA University, Mid term Examination, Odd Semester 2022-23

r_k	0	1	2	3	4	5	6	7
n_k	8	10	10	2	12	16	4	2

Target Image Histogram

r_k	0	1	2	3	4	5	6	7
n_k	13	12	14	14	11	0	0	0

GLA UNIVERSITY MATHURA Accredited with A+ Grade by NAAC 12-B Status from UGC

Numerical Examples

Ques: A and B are two 3-bit image. Histogram Equalisation was done on both the images and following results were obtained.

GLA University, Mid term Examination, Odd Semester 2023-24

I/P gray level of Image A	0	1	2	3	4	5	6	7
O/P gray level of Image A	1	3	5	6	6	7	7	7

I/P gray level of Image A	0	1	2	3	4	5	6	7
O/P gray level of Image A	0	0	0	1	2	5	6	7

Numerical Examples

We want to match the histogram of Image A to that of Image B. What will be the result of histogram matching? Give your answer in the following format.

I/P gray level of Image A	0	1	2	3	4	5	6	7
O/P gray level of Image A after matching								

Numerical Examples

Ques1: Suppose that a 3-bit image of size 64 × 64 pixels has the intensity distribution shown in the table (on the left). Get the histogram transformation function and make the output image with the specified histogram, listed in the table on the right

TYPE 2

Table 1

r_k	n_k	$p_r(r_k) = n_k/MN$
$r_0 = 0$	790	0.19
$r_1 = 1$	1023	0.25
$r_2 = 2$	850	0.21
$r_3 = 3$	656	0.16
$r_4 = 4$	329	0.08
$r_5 = 5$	245	0.06
$r_6 = 6$	122	0.03
$r_7 = 7$	81	0.02

Table 2

z_q	Specified $p_z(z_q)$
$z_0 = 0$	0.00
$z_1 = 1$	0.00
$z_2 = 2$	0.00
$z_3 = 3$	0.15
$z_4 = 4$	0.20
$z_5 = 5$	0.30
$z_6 = 6$	0.20
$z_7 = 7$	0.15

Numerical Examples

Ques: histogram of 3-bit/pixel square image is as shown below:

GLA University, Mid term Examination, Even Semester 2023-24

r_k	0	1	2	3	4	5	6	7
n_k	4	4	0	0	0	3	2	3

Specified normal histogram (pdf) is as shown below

r_k	0	1	2	3	4	5	6	7
pdf	0.15	0	3	0	0.15	0	0.4	0

Numerical Examples

Ques 3: Histogram of a 3-bits/pixel square image is given in figure 1,.

GLA University, Mid Term Examination, Odd Semester 2024-25

TYPE 3

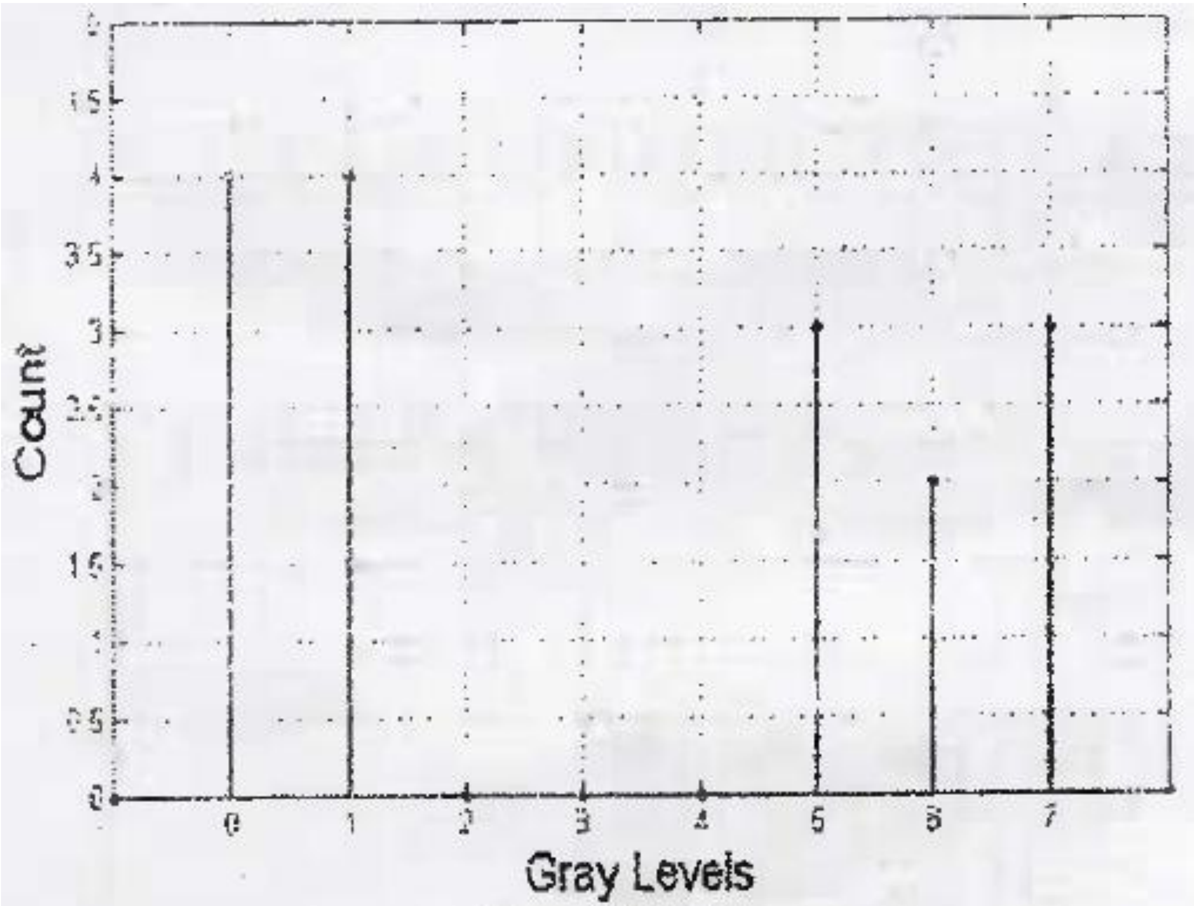


Figure-1:Histogram

Numerical Examples

TYPE 3

specified normalised histogram (pdf) is given in figure 2. Use histogram specification to generate histogram of enhanced image

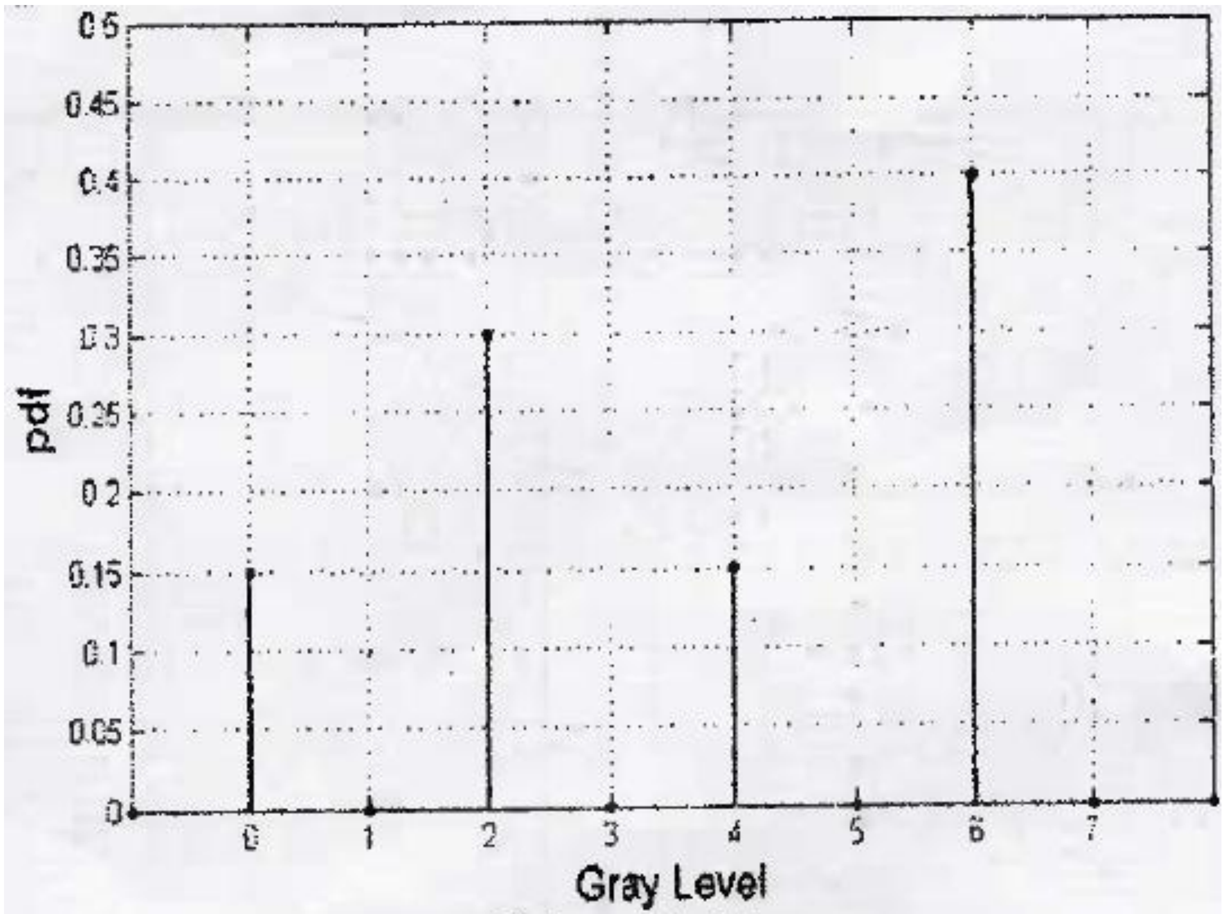


Figure 2: Specified normalized histogram

Histogram Stretching

Numerical Examples

TYPE 1

Ques1: Perform histogram stretching on the 8 × 8, eight level gray image, the gray level distribution of which is show in table. Perform the histogram stretching from gray level 0 to 7

r_k	0	1	2	3	4	5	6	7
n_k	0	0	5	20	20	19	0	0

GLA UNIVERSITY MATHURA
Accredited with A+ Grade by NAAC
12-B Status from UGC

Numerical Examples

TYPE 2

Ques2: A certain image has 10 Gray level intensities in the range 6 to 15. If we generate a linear contrast stretched image with min and max Gray level are 0 to 7 respectively. Write the formula and show the mapping in the tabular format as shown below.

GLA University, Mid Sem Examination, 2023-24

r	s	Round Off

Numerical Examples

TYPE 3

Ques3: Assume a 4-bit gray scale I as shown below. Generate the linear contrast stretched image with minimum gray level 5 and maximum gray level 14

GLA University, End Sem Examination, 2021-22

3	3	3	3	3	3	3	3
3	4	4	4	4	4	4	3
3	4	2	2	2	2	4	3
3	4	2	5	5	2	4	3
3	4	2	5	5	2	4	3
3	4	2	2	2	2	4	3
3	4	4	4	4	4	4	3
3	3	3	3	3	3	3	3

Numerical Examples

TYPE 3

Ques: Convert the following image into a 4-bit image

GLA University, Mid term Examination, Even Semester 2022-23

255	200	170
130	100	70
50	30	10

Numerical Examples

TYPE 3

Ques: Consider the following image into a 3-bit image

GLA University, Mid term Examination, Odd Semester 2022-23

210	100	93
20	190	0
5	30	60

Histogram Statistics For Enhanced Image

GLA UNIVERSITY MATHURA
Accredited with A+ Grade by NAAC
12-B Status from UGC

Numerical Examples

Ques1: Consider the following 2-bit image of size 5×5

0	0	1	1	2
1	2	3	0	1
3	3	2	2	0
2	3	1	0	0
1	1	3	2	2

Determine the (A) Mean and (B) Variance, and (C) Standard Deviation.

Thank You